

Multiple Choice (10 marks)

1. Which of the following are the spatial clustering algorithms?
 - (a) Partitioning based clustering
 - (b) K-means clustering
 - (c) Grid based clustering
 - (d) All of the above
2. Another term for out-of-distribution detection is?
 - (a) anomaly detection
 - (b) one-class detection
 - (c) train-test mismatch robustness
 - (d) background detection
3. Which of the following points would Bayesians and frequentists disagree on?
 - (a) The use of a non-Gaussian noise model in probabilistic regression.
 - (b) The use of probabilistic modelling for regression.
 - (c) The use of prior distributions on the parameters in a probabilistic model.
 - (d) The use of class priors in Gaussian Discriminant Analysis.
4. Which of the following best describes the joint probability distribution $P(X, Y, Z)$ for the given Bayes net. $X \leftarrow Y \rightarrow Z$?
 - (a) $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$
 - (b) $P(X, Y, Z) = P(X) * P(Y|X) * P(Z|Y)$
 - (c) $P(X, Y, Z) = P(Z) * P(X|Z) * P(Y|Z)$
 - (d) $P(X, Y, Z) = P(X) * P(Y) * P(Z)$
5. Which of the following best describes what discriminative approaches try to model? (w are the parameters in the model)
 - (a) $p(y|x, w)$
 - (b) $p(y, x)$
 - (c) $p(w|x, w)$
 - (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: Clustering is effectively used for detecting fraudulent credit card transactions by grouping similar transaction patterns.
7. True or False: The rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is 1 due to its linearly dependent rows.
8. True or False: The Penn Treebank is primarily focused on sentiment analysis and does not include language modeling applications.
9. True or False: The dimensionality of the null space of the matrix A is 2.
10. True or False: In the given Bayesian network, 15 independent parameters are required when no assumptions about independence are made.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to extract maximum and minimum k elements in the given tuple.
12. Write a function to sort a given list of elements in ascending order using heap queue algorithm.

End of Paper